

# **Explain what shape memory alloys are and how they are used in engineering applications**

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Course: *MIET2115*

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## Introduction

Shape-Memory Alloys (SMAs) are a class of material which when exposed to a stimulus, most commonly heat, will return to a previously memorised shape. They were first discovered in 1932 by Arne Ölander with the first alloy using gold-cadmium, however, nitinol (NiTi), the most used

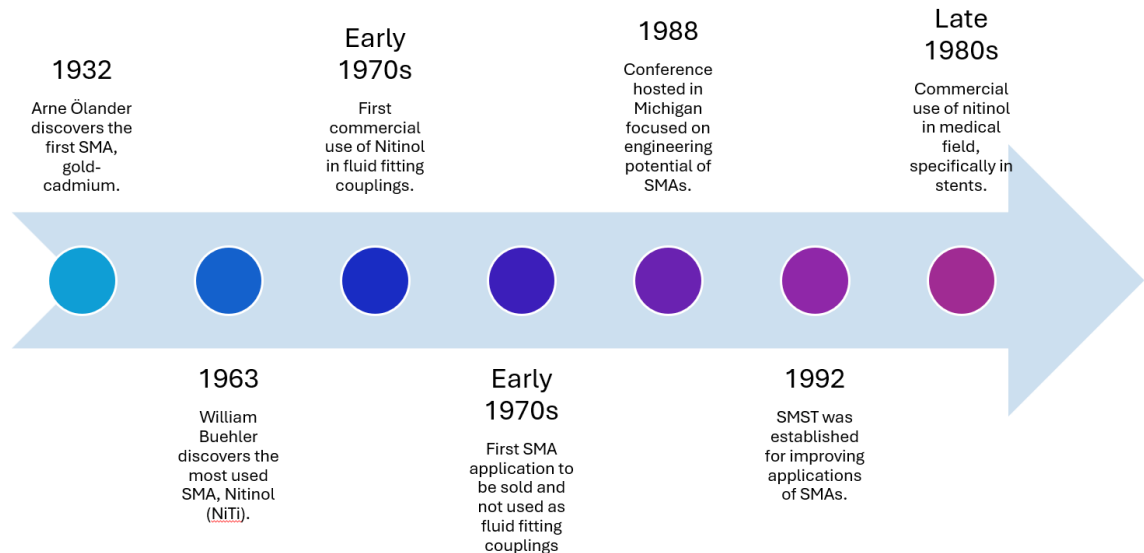


Figure 1: SMA historical breakthroughs. Adapted from (Sparks & Falkowski 2019), (AccuPath 2025), (Duerig & Pelton n.d.) and (Wikipedia contributors 2026)

SMA was discovered in 1963 by William Buehler when testing different materials for a missile nose tip. SMAs are becoming increasingly prevalent in medical, aerospace and civil engineering fields with its unique properties offering a simplistic and reliable alternative to conventional mechanical devices. This report will explore how SMAs function as well as their creation, applications and additional considerations for engineers.

## What are Shape Memory Alloys?

### Definition

L. McDonald Schetky's report titled Shape-Memory Alloys, "If [Shape Memory Alloys] are plastically deformed at one temperature, they will completely recover their original shape on being raised to a higher temperature." (Schetky 1979, p. 74)

### Key Features

Materials that are considered shape memory alloys (SMA) retain thermomechanical treatments that they have undergone. Christian LExcellent states that the SMA's "memory is developed by way of training – i.e. often by the repetition of the same thermodynamic loading... in terms of stress or strain imposed and/or in terms of temperature." (LExcellent 2013, ch. 1)

### Common Shape Memory Alloy Materials

The first shape memory alloy discovered by Arne Ölander in 1932 was in a gold-cadmium alloy. Nitinol (Ni-Ti) is an SMA that is made up of roughly equal portions of nickel and titanium. This 50/50 ratio is important as if this changes it will change the temperature where the phase

transformation occurs. Nitinol is the default alloy when thinking of SMAs, however there are many different materials which can make up an SMA. Some include copper-aluminium-nickel and nickel-titanium, which has already been covered above. The different alloys allow for engineers to tailor the SMAs to best fit their purpose.

## Underlying Mechanisms

Shape-Memory Alloys have multiple stable phases at different temperatures. When changing between these solid-state phases, the crystalline structure is rearranged, but the material remains as a solid. This is a displacive transformation as the change in phase happens without diffusion through a coordinated movement of groups of atoms relative to their neighbours. The two phases of SMAs are martensitic and austenitic. The martensitic phase is also known as the super plastic phase, when transitioning to the austenitic phase, also known as the super elastic phase, the alloy will revert to the memorised shape.

Additionally, Shape-Memory Alloys have very high fatigue resistance. When bending occurs in the martensitic stage, like any material defects occur, but during the transition to the austenitic phase, these defects are destroyed. This assumes 100% of the alloy is transitioned, as sections are not converted the defects will be present in the austenitic phase. X and the other authors on this paper state that “The superelasticity in Ni-Ti [nitinol] can recover from about 8-10% strain, before it deforms plastically.” (Cherain Chekotu et al. 2022, sec. 1. Introduction)

## Manufacturing

The process of creating shape-memory alloys is essential for the material to fulfill its purpose. As an example, a paper titled ‘Control of Ni-Ti phase structure, solid-state transformation temperatures and enthalpies via control of L-PBF process parameters’ tested the effect of changing the ratio between nickel and titanium and its correlation with altered temperatures which a phase change occurs. The paper used a base composition of 49.9% nickel and 50.1% titanium. The authors state that “Ni evaporation or increase in Ti content in the matrix have a significant effect on the shape memory behaviour of the material by increasing the [transformation temperatures].” (Cherain Chekotu et al. 2022, sec. 3.5 Nickel Evaporation) Altering the makeup of the alloy can be a tool for engineers to utilise to optimise the shape-memory alloy for its purpose.

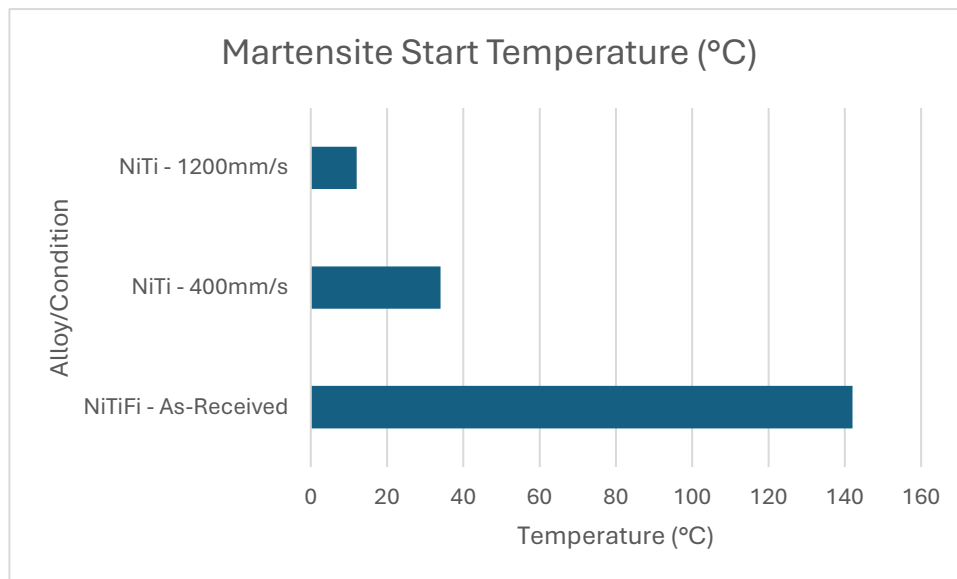


Figure 2: Difference in martensite start temperature varies with different alloys/conditions. Adapted from (Zhang et al. 2022), (Wang et al. n.d.)

## Applications in Engineering

Shape-memory alloys have many current engineering applications, such as strengthening existing concrete, rehabilitation of damaged concrete and for seismic resilient building materials due to its super elasticity. They are most established in the medical industry, being used in devices such as vascular stents. In their paper ‘Stainless and shape memory alloy coronary stents: a computational study on the interaction with the vascular wall’, Francesco Migliavacca and his fellow authors claim shape memory alloy stents “induces fewer stresses and lower damage to the vessel” (Migliavacca et al. 2004, sec. Abstract) when compared to stainless steel stents. This is due to their ability to endure large strains when compressing to fit into the body.

Shape-memory alloys are also used in aerospace applications as it offers a simpler alternative to mechanical systems in areas which face harsh pressures and temperatures. This property also makes SMAs attractive for uses in oil, gas and energy systems where they face similar extreme situations.

Another fortuitous use for SMAs has been in robotics, such as limb devices. This comes in the form of shape-memory alloy actuators. SMA wires and springs are installed to mimic muscles and they operate by inducing changes in the phase, which cause the wires and springs to expand and contract, generating the force that makes movement in the ‘limb’.

## Other Considerations

A study ‘Sustainability and resilience of steel – shape memory alloy reinforced concrete bridge under compound earthquakes and functional deterioration within entire life-cycle’ “confirm[s] that the steel-SMA reinforced concrete bridge is more sustainable and resilient under earthquakes and functional deterioration in a long-term perspective” (Qian et al. 2022, sec. Abstract). It is also concluded in comparison to a regular bridge that while shape-memory alloy

bridges produce slightly greater amounts of carbon dioxide during construction, it is more corrosion resistant therefore having a longer lifespan. However, it predicts that over the entire lifespan, it will have a far smaller carbon footprint than a traditional bridge due to less damage, so therefore less repairs and less consequential carbon emissions from external factors such as increased vehicle travel time. So, while shape-memory alloy bridges produce heightened carbon emissions during construction, over its lifespan it will reduce the overall emissions relative to other bridges.

A drawback with SMAs described in the paper ‘Shape memory alloys in modern engineering: progress, problems, and prospects’ is having “exorbitant manufacturing cost[s]” (Hossain, Rabbi & Ali 2025, sec. 1. Introduction). Prices for 1kg of 0.5mm nitinol wire is \$2406.00 AUD (“Nitinol SMA Shape Memory Alloy Wire” n.d.), which is a serious affordability issue for products which use this or any other SMA. A potential solution to this issue is to use alternative copper or iron based SMAs, which are cheaper to produce, however have lower recovery strains and can oxidise.



Figure 3: Price differences between 1kg of 0.5mm alloy wire. Adapted from (“Nitinol SMA Shape Memory Alloy Wire” n.d.), (“DN2E0,50-1KG INDEL - Coil wire” n.d.), (Kelvin Trading Pty Ltd (Beekeeping Supplies) 2024), (“0.5mm Enamel Copper Wire Spool” n.d.)

## Conclusion

This report will explore how SMAs function as well as their creation, applications and additional considerations for engineers. SMAs switch between martensitic and austenitic phases, achieving a repeatable actuation without the use of moving parts which are more prone to failure. Their superelasticity also bolsters their use over mechanical systems, being able to withstand extensive pressures before failure, as well as their ability to ‘heal’ defects that are

incurred. High costs and hard manufacturability offer further room for further advancement in SMA use, giving the industry a vast area for growth.

## Appendix

### Appendix A – AI Prompt Refinement

- Initial prompts were broad and gave vague answers from Val. (*How do shape-memory alloys change.*)
- The next change was to give clearer instructions for an answer pinpointed on the details I want to know. (*Explain and outline how shape-memory alloys change properties and shape when subject to temperature change and forces. In your response, describe the change between martensite and austenite phases and how each phase effects the alloys properties such as elasticity and defects as well as the crystal structure.*)
- Next constraints were added to optimise the response, ridding the AI's answer of unnecessary jargon. (*Explain and outline how shape-memory alloys change properties and shape when subject to temperature change and forces. In your response, describe the change between martensite and austenite phases and how each phase effects the alloys properties such as elasticity and defects as well as the crystal structure. Limit the response to using only clear and simple engineering language and avoid unneeded jargon and filler words.*)

### Answer to Original Prompt

Chatbot used: claude

#### Shape-Memory Alloys (SMAs): How They Change

Shape-memory alloys are remarkable materials that can "remember" a previous shape and return to it when stimulated. Their behavior stems from a solid-state phase transformation between two crystal structures:

#### The Two Phases

- **Austenite (parent phase)** — stable at *higher temperatures*. The crystal structure is ordered and symmetric (typically cubic). This is the "remembered" shape.
- **Martensite (product phase)** — stable at *lower temperatures*. The structure is less symmetric and can be mechanically deformed without permanently damaging the atomic lattice.

#### The Transformation

When an SMA cools below a critical temperature, it transforms from austenite into martensite. In this martensitic state, the alloy is relatively soft and can be bent, stretched, or compressed. Crucially, this deformation doesn't break atomic bonds — it only reorients the crystal variants (called **twinning**).

When reheated above the transformation temperature, the crystal reverts to the austenite structure, and the material snaps back to its original shape — sometimes with considerable force.

#### Types of Memory Effect

| Effect          | Description                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One-way effect  | Remembers one shape; returns to it only when heated                                                                                                            |
| Two-way effect  | Remembers two shapes — one for hot, one for cold — switching between them with temperature                                                                     |
| Superelasticity | At certain temperatures, deformation and recovery happen <i>isothermally</i> (no heating needed) — the material springs back immediately upon releasing stress |

### Key Variables That Drive the Change

- **Temperature** — the primary trigger; the transformation occurs across a specific range (not a single point), defined by four temperatures: Ms, Mf, As, Af
- **Stress** — applied mechanical stress shifts the transformation temperatures (higher stress = higher transformation temp)
- **Composition** — alloying ratios directly control the transformation temperature range

### Common Alloys

- **Nitinol (Ni-Ti)** — the most widely used; biocompatible, used in stents, orthodontic wires, and actuators
- **Cu-Zn-Al** and **Cu-Al-Ni** — cheaper alternatives, more brittle
- **Fe-Mn-Si** — used in engineering fasteners and pipe couplings

### Why It Works at the Atomic Level

The transformation is *diffusionless* — atoms don't swap places, they shift cooperatively by tiny, coordinated movements (a shear-like distortion). This makes the transformation very fast and highly repeatable, allowing SMAs to cycle hundreds of thousands of times without fatigue in well-designed applications.

## Answer to Improved Prompt

Chatbot used: *claude*

### Shape-Memory Alloys: Phase Transformation and Mechanical Behaviour

#### What Shape-Memory Alloys Do

Shape-memory alloys (SMAs) can be deformed at low temperature, then return to their original shape when heated. This behaviour is driven entirely by a reversible change in crystal structure between two solid phases: **martensite** (low temperature) and **austenite** (high temperature).

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### The Two Phases

#### Austenite — High-Temperature Phase

- Has a **cubic crystal structure** (body-centred cubic in most Nitinol-type alloys).

- The lattice is highly ordered and symmetric.
- The alloy is **stiff and strong** — high Young's modulus.
- This is the **parent phase** and defines the alloy's "remembered" shape.

### Martensite — Low-Temperature Phase

- Has a **monoclinic or orthorhombic crystal structure** — lower symmetry, distorted lattice.
- Forms from austenite by a **diffusionless shear transformation**: atoms shift cooperatively by small distances without any compositional change.
- The alloy is **soft and easily deformed** — low yield stress, lower Young's modulus.
- Deformation in martensite occurs not by dislocation slip (as in ordinary metals) but by **twin boundary movement**, which is fully reversible.

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### The Transformation Sequence

Four temperatures define the transformation:

#### Temperature Event

|                      |                                               |
|----------------------|-----------------------------------------------|
| <b>M<sub>s</sub></b> | Martensite starts forming on cooling          |
| <b>M<sub>f</sub></b> | Martensite formation complete                 |
| <b>A<sub>s</sub></b> | Austenite starts forming on heating           |
| <b>A<sub>f</sub></b> | Austenite formation complete, shape recovered |

There is always **thermal hysteresis** between the cooling and heating paths — the alloy does not transform at the same temperature in both directions.

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### How Deformation Works in Each Phase

#### In Martensite (the "deformable" state)

When stress is applied, the martensite accommodates strain by reorienting its twin variants. Twin boundaries — the internal interfaces between differently oriented lattice regions — move under low stress. This produces **large apparent strains (up to ~8%)** without creating permanent defects such as dislocations. Because no dislocations are generated, the deformation is **fully recoverable** once heat is applied.

#### In Austenite (the "stiff" state)

Austenite resists deformation. If stress is applied above the transformation temperature, the alloy can also be forced from austenite directly into martensite — this is the **superelastic (pseudoelastic) effect**. When the stress is removed, the martensite is unstable at that temperature and reverts to austenite, recovering the strain with no heating required.

### Crystal Defects

- **Twin boundaries** in martensite are the key structural feature — they are mobile, low-energy interfaces that allow reversible shape change. These are not damaging defects.
- **Dislocations** are the enemy of shape-memory behaviour. If stress is too high or cycling is excessive, irreversible dislocation slip occurs, which degrades the memory effect over time.
- The transformation itself generates **internal stresses** at grain boundaries due to the volume and shape change of the lattice — this is why grain size and alloy composition must be carefully controlled.

### Summary of Property Changes by Phase

| Property                  | Martensite                               | Austenite                                    |
|---------------------------|------------------------------------------|----------------------------------------------|
| Crystal structure         | Monoclinic / orthorhombic (low symmetry) | Cubic (high symmetry)                        |
| Young's modulus           | ~30–40 GPa (Nitinol)                     | ~75–83 GPa (Nitinol)                         |
| Yield stress              | Low (~100–200 MPa)                       | High (~400–700 MPa)                          |
| Deformation mechanism     | Twin boundary motion                     | Dislocation slip / stress-induced martensite |
| Deformation reversibility | Fully reversible                         | Reversible (superelastic regime)             |
| Temperature               | Below $M_f$                              | Above $A_f$                                  |

### The Shape-Memory Cycle in Brief

1. **Start in austenite** — alloy holds its set shape, stiff.
2. **Cool below  $M_f$**  — transforms to martensite, becomes compliant.
3. **Apply load** — twin boundaries reorient; alloy deforms at low stress.
4. **Remove load** — deformed shape is retained (martensite is stable at low temperature).
5. **Heat above  $A_f$**  — martensite reverts to austenite; original shape is recovered as the crystal structure returns to cubic symmetry.

## Prompt Refinement Ranking

- The most effective technique was to instruct the AI to use engineering language and not include any filler sentences. It conveyed a more concise response.
- The next was to outline sections of information that would be useful to me so I don't receive a response with sections that are unrelated to this report.
- Chaining these responses was the least important as I found that the language model would get stuck up on regurgitating previous responses in a different format rather than outputs being entirely new. This can easily be avoid through using a different chats for each new prompt.

## Appendix B – Database Search Strategy

| DATABASE       | SEARCH STRING                                                                 |
|----------------|-------------------------------------------------------------------------------|
| Google Scholar | “Shape-memory alloy” AND “Key features”                                       |
| Google Scholar | “Shape-memory alloy” AND “Material makeup”                                    |
| Google Scholar | “Shape-memory alloy” AND (Aerospace OR Civil OR Mechanical) AND “Engineering” |
| Google Scholar | “Shape-memory alloy AND Stent*                                                |
| Google         | “Shape-memory alloy” AND “limb devices”                                       |
| Google         | “Shape-memory alloy” AND (Cost* OR Sustainability*)                           |
| Google         | “Shape-memory alloys” AND Applications*                                       |
| Google         | Nitinol or “NiTi” AND (Price* OR Cost*)                                       |
| Google Scholar | “Shape-memory alloy” AND (Issue* OR Problem* OR Challenge*)                   |

## Appendix C – Data Verification

| Cycle | Source Origin                                   | Source                                                               | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|-------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Report                                          | (Cherain Chekotu et al. 2022, sec. 1. Introduction)                  | “Ni-Ti [nitinol] can recover from about 8-10% strain, before it deforms plastically.”                                                                                                                                                                                                                                                                                                                                            |
| 2     | Claude (AI)                                     | Prompt                                                               | “Nitinol (nickel-titanium shape memory alloy) can withstand strains of up to <b>8–10%</b> before undergoing permanent plastic deformation”                                                                                                                                                                                                                                                                                       |
| 3     | Website (Lawrence Berkeley National Laboratory) | ( <i>Mechanical fatigue and fracture of Nitinol</i> 2012b, p. 1)     | “Nitinol displays recoverable strains... as high as 10%.”                                                                                                                                                                                                                                                                                                                                                                        |
| 4     | Claude Follow Up (AI)                           | Refined prompt to be more accurate and scout for any contradictions. | “Nitinol can superelastically recover strains of up to <b>~8%</b> without any permanent deformation, but irreversible plastic deformation begins at approximately <b>10–12% total strain</b> — a figure consistent with my previous answer and carrying no contradiction, since the 8–10% range cited earlier refers to the recoverable superelastic regime, which sits safely below the ~10–12% plastic deformation threshold.” |

| Cycle | Source Origin          | Source                                                              | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Report                 | (Cherain Chekotu et al. 2022, sec. 3.5 Nickel Evaporation)          | “Ni evaporation or increase in Ti content in the matrix have a significant effect on the shape memory behaviour of the material by increasing the [transformation temperatures].”                                                                                                                                                                                                                                                                                                                                |
| 2     | Report                 | (Ratajski et al. 2024, sec. Conclusion)                             | “The transformation temperature M(B19') → A(B2) increases with the ageing time.”                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3     | ChatGPT (AI)           | Prompt                                                              | “Yes—changes in the NiTi matrix (such as composition, defects, precipitates, or internal stress) shift the transformation temperatures because they alter the stability of the austenite and martensite phases.”                                                                                                                                                                                                                                                                                                 |
| 4     | ChatGPT Follow Up (AI) | Refined prompt to be more accurate and check for any contradictions | “Altering the NiTi compositional matrix—especially the Ni:Ti ratio, precipitate content, or alloying additions—does shift the martensitic transformation temperatures (Ms, Mf, As, Af) by changing phase stability, with the key nuance that Ni-rich compositions generally lower these temperatures while Ti-rich or certain alloying additions can raise them, though the exact effect can vary depending on microstructure, processing history, and whether intermediate phases (e.g., R-phase) are present.” |

## Appendix D – Teamwork Analysis

The teamwork in this report was not applicable. Having a team to work with on this project would ensure that a greater number of sources were analysed leading to increased accuracy as well as having more diverse expert perspectives on the topic. In joining a team, skills that are transferable to other group projects and into the workplace could’ve been improved.

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# AI Declaration

## AI Declaration Form

Course: MIET2115 – Mechanics and Materials 2

Hugh Williams / ██████

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### 1. Declaration of AI Use

*AI tools include (but are not limited to): VAL, ChatGPT, Claude, GitHub Copilot, Grammarly, DeepSeek, Bing/Google AI features, or any software/online service that generates or edits text, code, images, or design outputs.*

Please select one option below:

- ☐ **No AI Use:** We did not use any AI tools in preparing this report.
- ☐ **Language/Editing Support Only:** We/I used AI tools only for grammar, spelling, punctuation, formatting, or general language clarity. No technical, analytical, or design content was generated using AI.
- ☐ **AI Used for Understanding or Early Drafting:** We/I used AI tools to assist understanding, brainstorm ideas, or generate preliminary drafts (e.g., explanations, pseudocode, design concepts). All AI-assisted content was independently checked, verified, and rewritten in our own words.
- ☒ **Other AI Use:** We/I used AI tools in the Assignment 1 as per the related marking rubric

AI was used when instructed for data verification and prompt refinement techniques. It wasn't used in any metric in the generation of the report body. Additionally, it was used to scrape through pdfs of papers uploaded to it to get more quickly locate data found within.

### 2. Description of AI Use (if applicable)

If any AI use was selected above, briefly describe what was used and how it contributed to your work (e.g., clarifying theory, debugging hints, summarising articles, formatting tables, generating images, refining explanations, etc.).

**Example:** “We used ChatGPT to rephrase parts of the static analysis section for clarity. All calculations, interpretations, and final explanations were completed by us.”

AI was used when instructed for data verification and prompt refinement techniques. It wasn't used in any metric in the generation of the report body. Additionally, it was used to scrape through pdfs of papers uploaded to it to get more quickly locate data found within.

### 3. Verification Statement

By signing this form, we/I confirm that:

- All technical analysis, calculations, design decisions, and interpretations in the submission are our/my own work and based on our/my own understanding.
- Any AI-generated ideas or suggestions have been critically evaluated, verified, and edited by us/me.
- No AI tool was used to produce final technical results, code, or design outputs unless explicitly permitted in the assessment instructions.
- This submission complies with the academic integrity requirements of this course and the University.
- If there is any doubt as to the originality of the work, we/I can be interviewed by the course coordinator to determine our learning outcomes.

#### Signature

Team Representative / Individual Student Name: Hugh Williams / ██████

Signature:

x  \_\_\_\_\_

Date: 26/03/2026

*Note: For group submissions, the representative signs on behalf of the group and is responsible for ensuring all members agree with the declaration.*